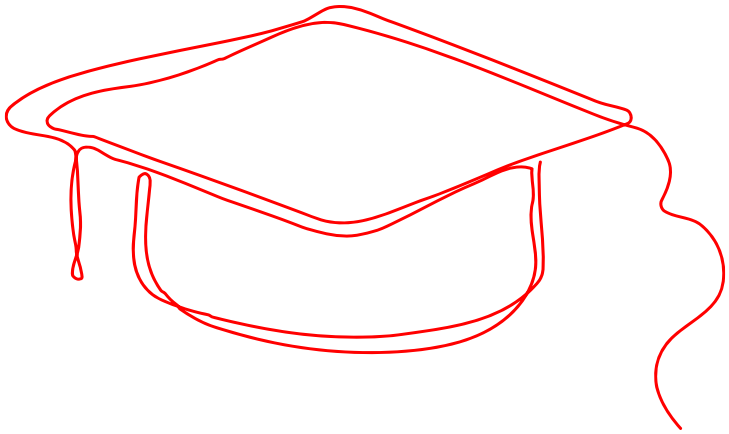
OPEN DAY 2026

CVC
Computer Vision Center

Vine a visitar els laboratoris, veure recerca en directe, conèixer investigadors i provar demostracions tecnològiques interactives!

REGISTRA'T AQUÍ →

7/05 | 11:00 – 12:30 h
15:00 – 16:30 h



Centre de Visió per Computador.
Edifici O, Campus UAB, Bellaterra

Neural Networks and Deep Learning

Guide to working on the project

Ramon Baldrich (ramon.Baldrich@uab.cat)
Credit to Dimosthenis Karatzas

Prerequisites

Prerequisites

Download and install

Visual Studio Code
(https://code.visualstudio.com/)

Create Account (and remember your credentials)

GitHub
(https://github.com/)

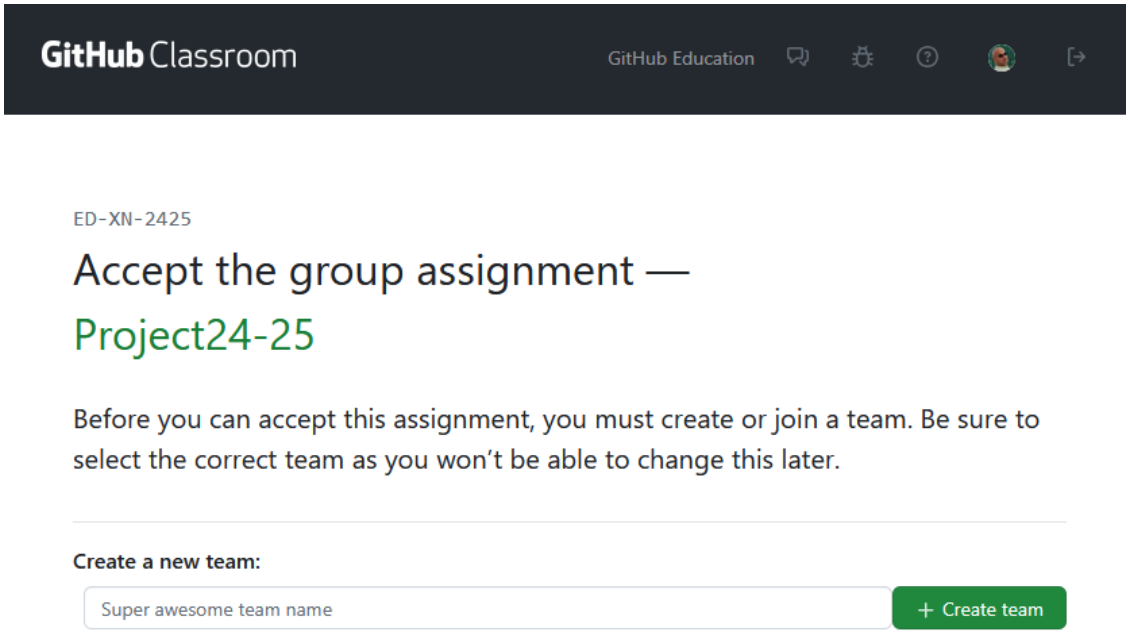
Weights and Biases
(https://wandb.ai/site)

Create a GitHub project for your team

Join, as a team, the assignment on the GitHub classroom of this class

https://classroom.github.com/a/CTxc2OD4

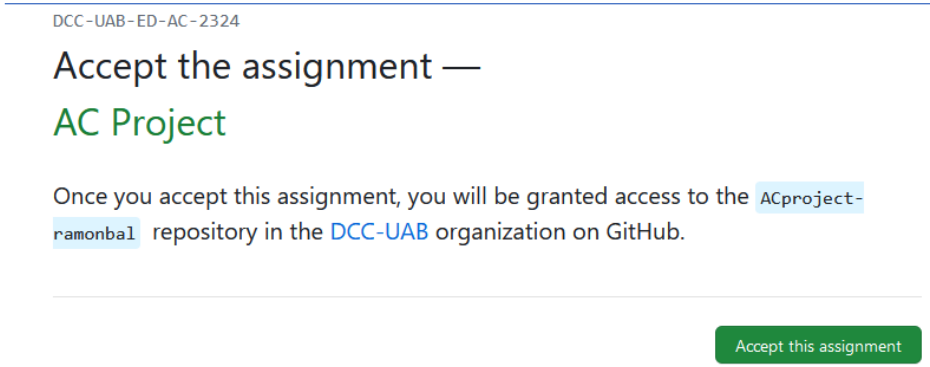
<- You should use this invitation link to join this assignment



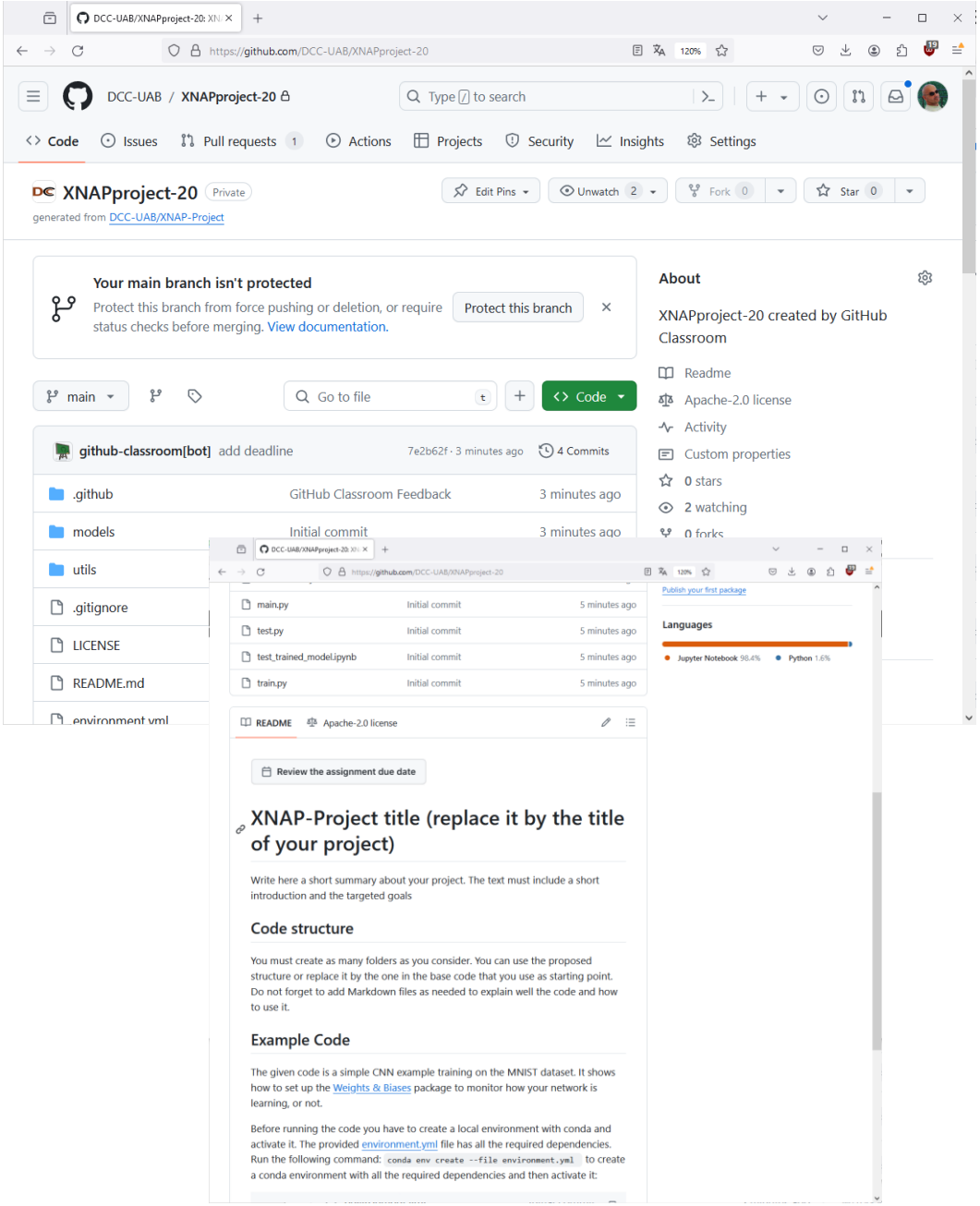
When you click on it, you are given the option to create a new team, or to join an existing one

IMPORTANT!

Only **ONE** of the members of your team should create a team and name it: "NN" where NN is your group id (01, …, 19)



Only then, the rest of the members should click on the invitation and **select your group team**



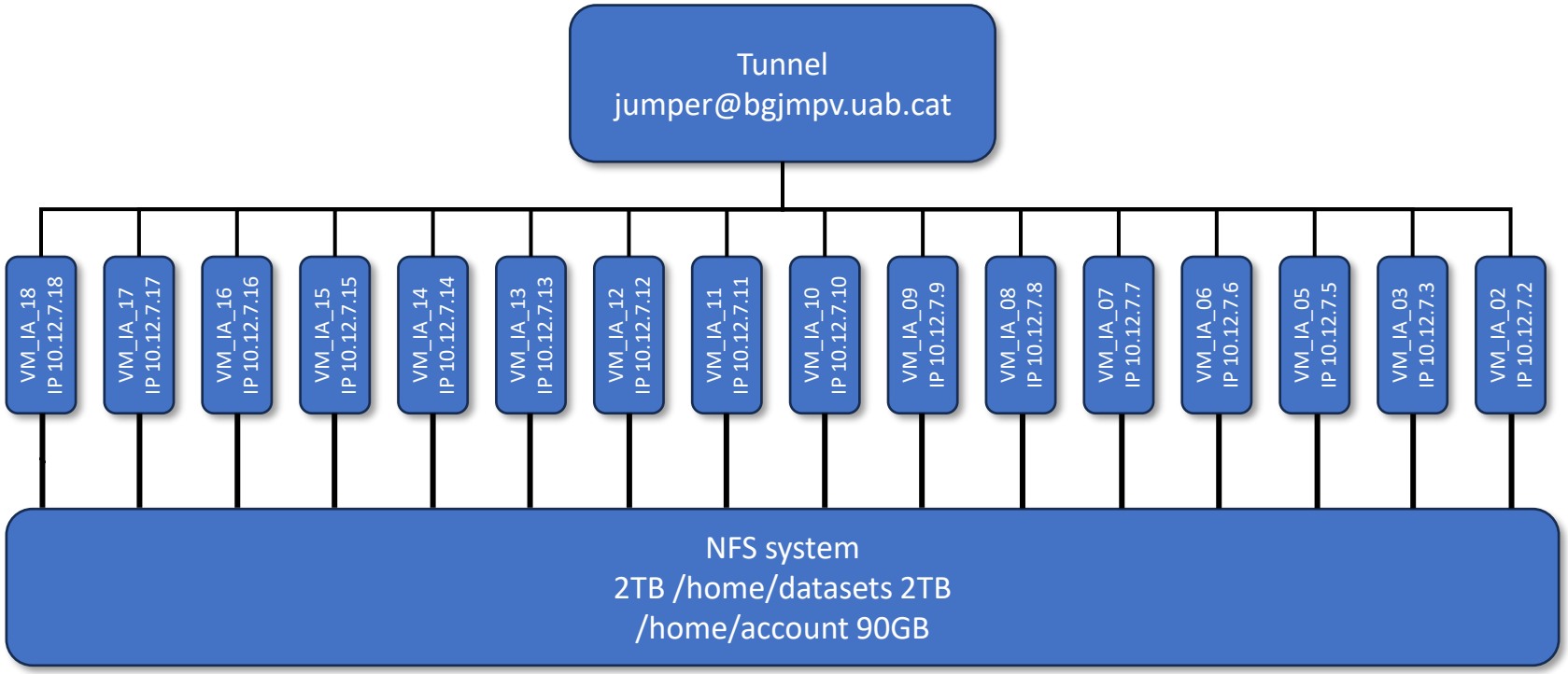
When you open the repository, you will see that you have a full example of a project there, that you can change to your own needs.

The example features a full structure of a working project, that trains a simple CNN on MNIST

UAB/DCC GPUs servers

ssh acces

ssh -J jumper@bgjmpvgpu.uab.cat group_account@10.12.7.X


Tunnel
jumper@bgjmpv.uab.cat
VM_IA_18 IP 10.12.7.18
VM_IA_17 IP 10.12.7.17
VM_IA_16 IP 10.12.7.16
VM_IA_15 IP 10.12.7.15
VM_IA_14 IP 10.12.7.14
VM_IA_13 IP 10.12.7.13
VM_IA_12 IP 10.12.7.12
VM_IA_11 IP 10.12.7.11
VM_IA_10 IP 10.12.7.10
VM_IA_09 IP 10.12.7.9
VM_IA_08 IP 10.12.7.8
VM_IA_07 IP 10.12.7.7
VM_IA_06 IP 10.12.7.6
VM_IA_05 IP 10.12.7.5
VM_IA_03 IP 10.12.7.3
VM_IA_02 IP 10.12.7.2
NFS system
2TB /home/datasets 2TB
/home/account 90GB

Access:

ssh -J jumper@bgjmpvgpu.uab.cat edxnGXX@10.12.7.X

nomarxxxx

edxnG00	Aqu7The3sh
edxnG01	ThieNae9ee
edxnG02	Eishoosh9d
edxnG03	chaeL9ohNg
edxnG04	weisoe3Too
edxnG05	aephuaX7ei
edxnG06	eequahzae7
edxnG07	ea4aequagh
edxnG08	peiquahng7
edxnG09	shieChihe9
edxnG10	yuCae3ohth
edxnG11	dohjoux3Ah
edxnG12	gai7yaenov
edxnG13	eebiegh4Je
edxnG14	Eejei7yoh9
edxnG15	eW9EexohPh
edxnG16	aeJ9Mox9th

Important:

A Virtual machine can hold multiple users. The GPU is shared among all users.

Hint:

Connect to a virtual machine with the lower users connected to its corresponding GPU. Check out the GPU memory usage

http://neptu.uab.es/GPU/dashboard.html

If there is any data that has to be shared among students, it will be located at:
/home/datasets

Basic instructions to start using GPU resources:

cat /home/datasets/install

```
bash ../datasets/Miniconda3-latest-Linux-x86_64.sh -b
~/miniconda3/bin/conda init bash

source ~/.bashrc

conda install -y -c conda-forge matplotlib tqdm wandb scipy

pip install torch torchvision --index-url
        https://download.pytorch.org/whl/cu126

pip install nvitop
```

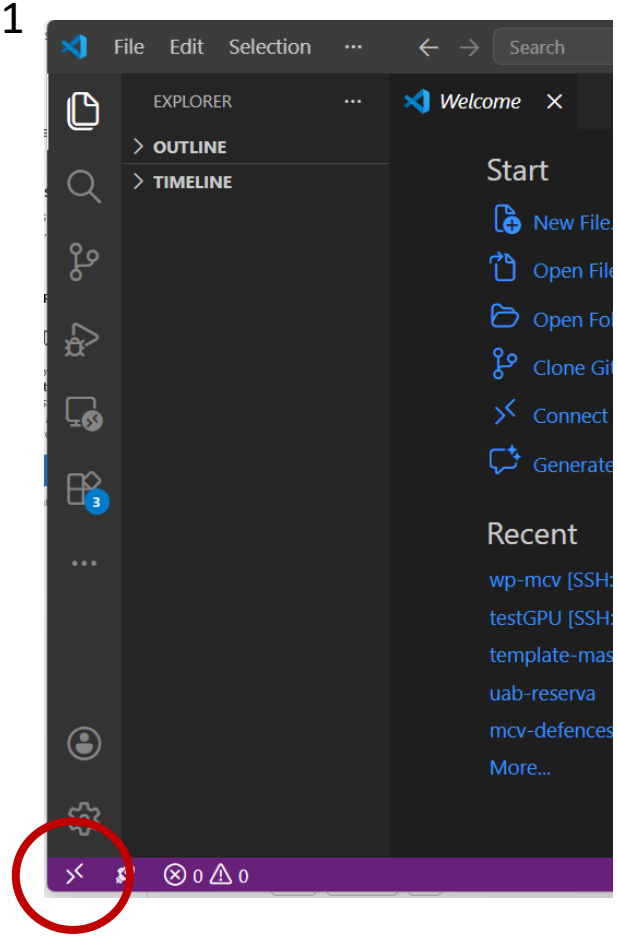
Using GitHub from VSCode in remote

Creating a repo on GitHub from a project at the remote server

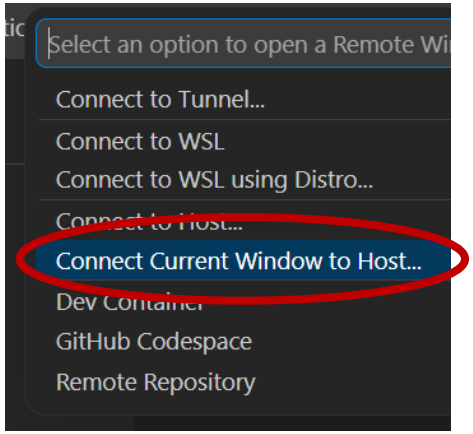
Connect to a remote server from VsCode

Adding remote servers into VsCode

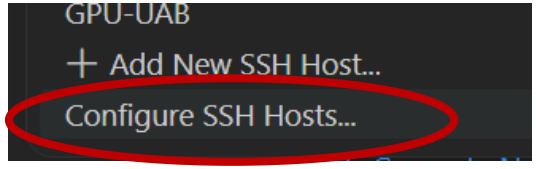
1



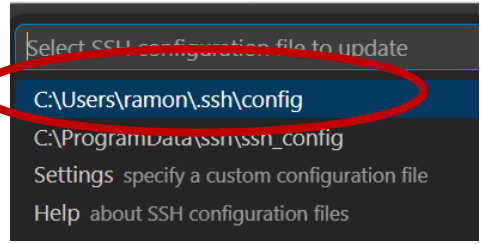
2



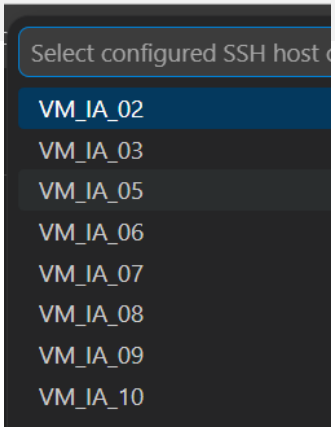
3



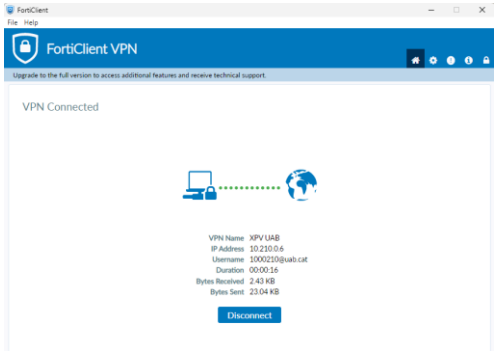
4



It opens a file, copy the content of file ‘config’ and save. If done correctly do steps 1 and 2 and you will get something like



Connecting to remote server with VsCode

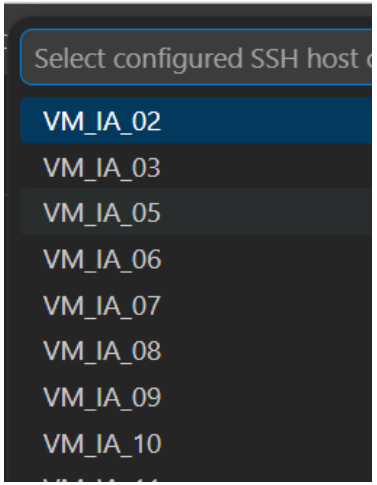


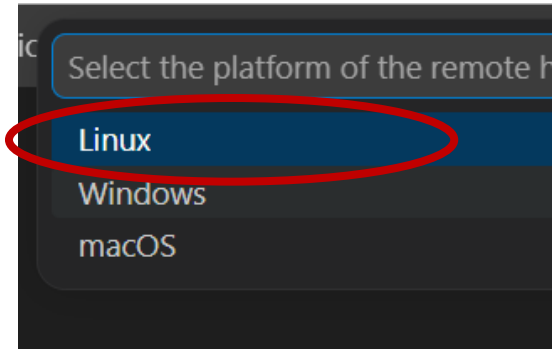
1 If you are out of UAB connect to a VPN with FortiClient

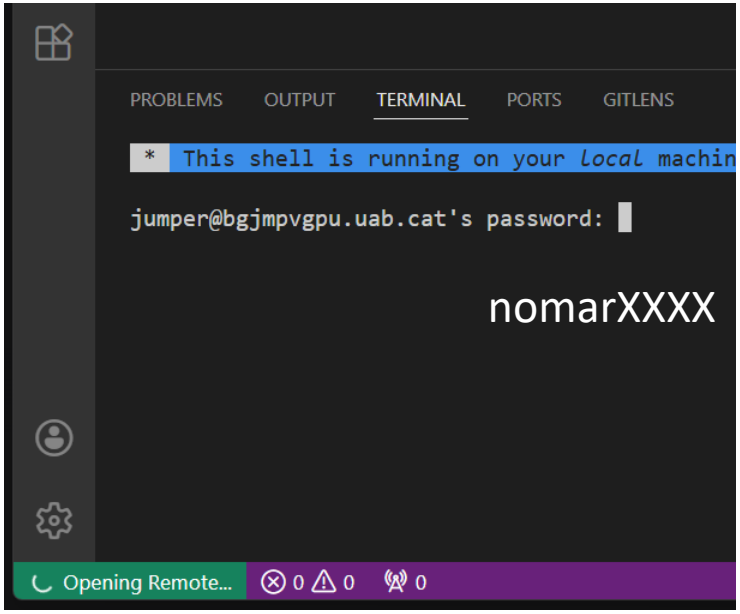
https://si-respostes.uab.cat/inici/teletreball/xarxa-privada-virtual

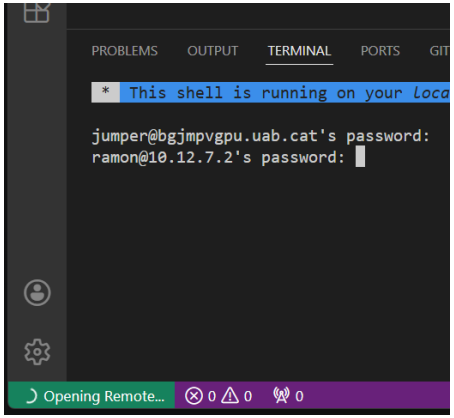
2 Select any of the servers (better choose one with low load)

http://neptu.uab.es/GPU/dashboard.html

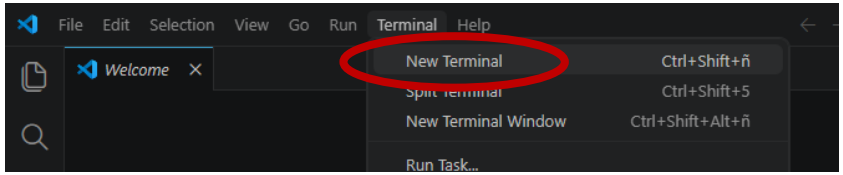




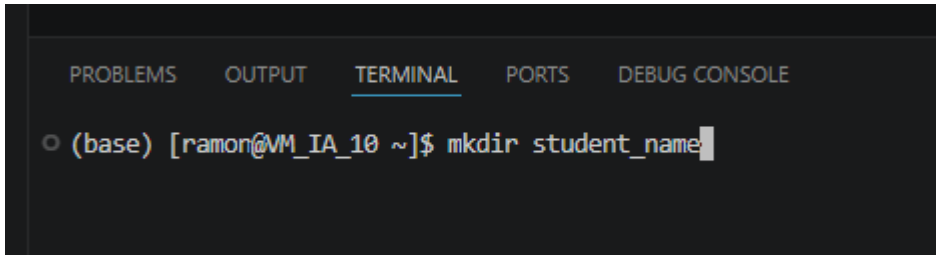




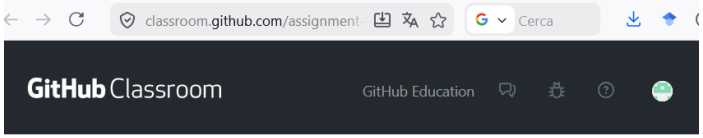
Clone your GitHub project on your virtual machine



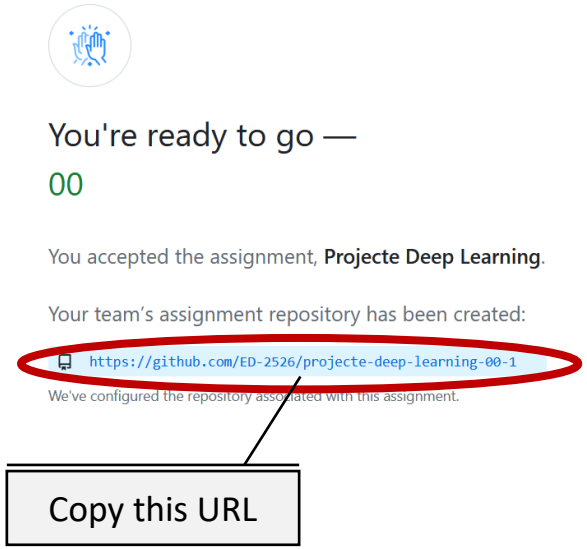
- Open a new terminal on the remote server
- Create a folder where to store our repo. Each student should have his own folder



https://classroom.github.com/a/CTxc2OD4



At your repo page on GitHub copy the URL given there



Then open a terminal window on VSCode, **go to your home directory**, and type:

```
git clone <URL>
```

Replacing <URL> with your copied url

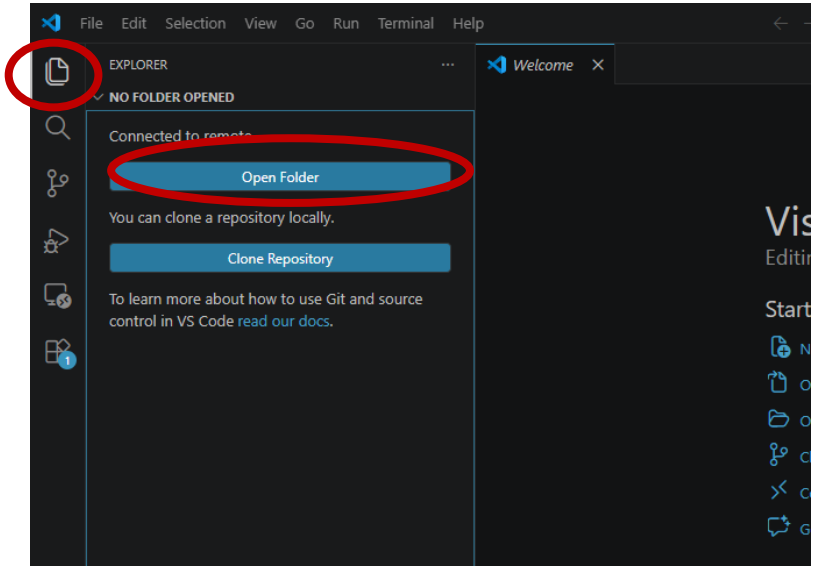
```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CONSOLE

(base) [ramon@VM_IA_10 ~]$ cd ramon
(base) [ramon@VM_IA_10 ramon]$ git clone  https://github.com/ED-2526/projecte-deep-learning-00-1
Cloning into 'projecte-deep-learning-00-1' ...
remote: Enumerating objects: 23, done.
remote: Counting objects: 100% (23/23), done.
remote: Compressing objects: 100% (20/20), done.
remote: Total 23 (delta 0), reused 20 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (23/23), 169.64 KiB | 1.31 MiB/s, done.
(base) [ramon@VM_IA_10 ramon]$
```

Setting your github account

Move to the folder with your repo and issue these two commands

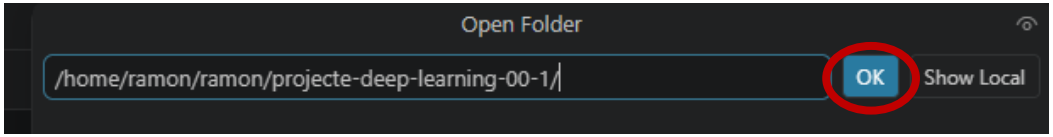
```
git config user.email "you@example.com"
git config user.name "Your Name"
```

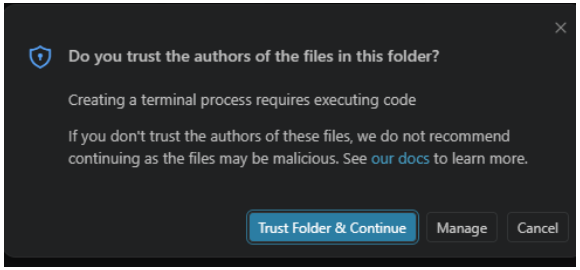
```
(base) [ramon@VM_IA_10 ramon]$ cd projecte-deep-learning-00-1/
(base) [ramon@VM_IA_10 projecte-deep-learning-00-1]$ git config user.email "ramonbc@gmail.com"
(base) [ramon@VM_IA_10 projecte-deep-learning-00-1]$ git config user.name  "ramon-test1"
(base) [ramon@VM_IA_10 projecte-deep-learning-00-1]$
```

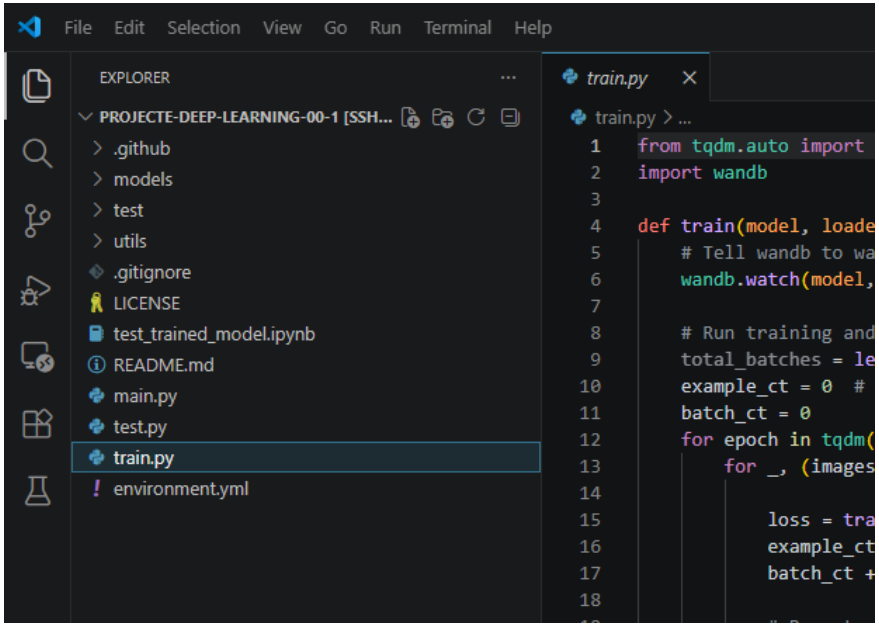


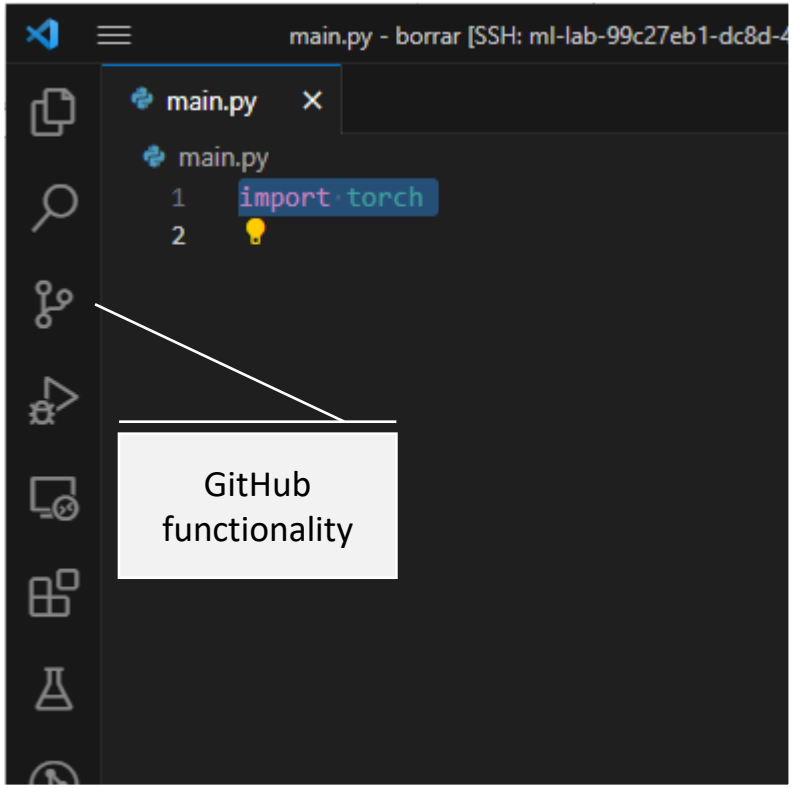
You should now see the cloned repository in a local folder of the same name, you be able to open it

You will be asked to log in again.



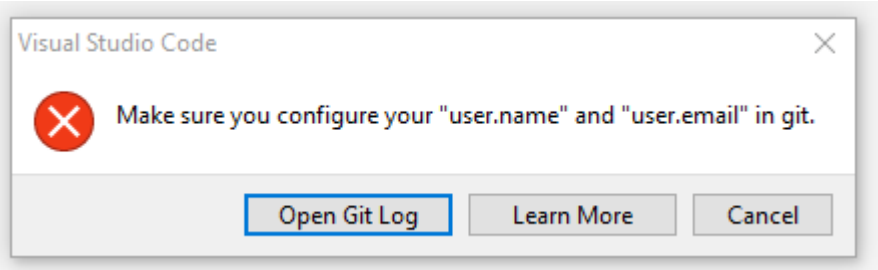




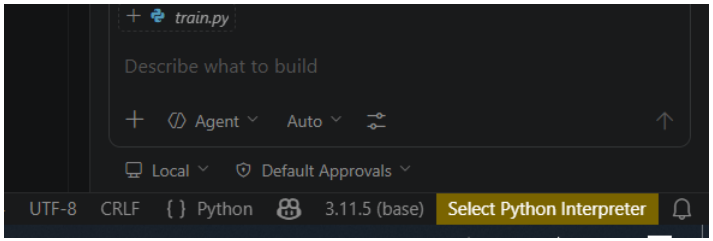


GitHub functionality is integrated in Visual Studio Code and you can access it from the side bar

But before you do so, you need to configure git on the virtual machine with your GitHub information

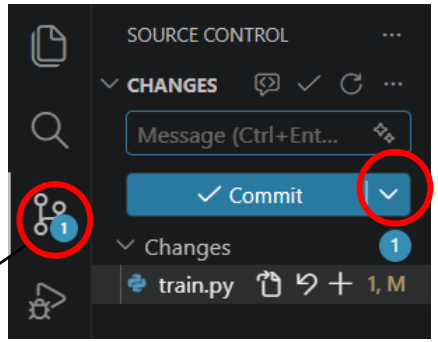


<- Otherwise, you will get an error like this when you try to use Git functionality

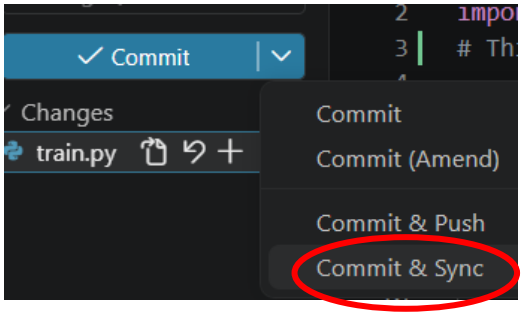


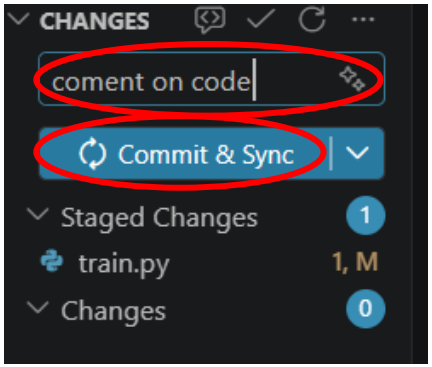
How to deploy changes to your mates

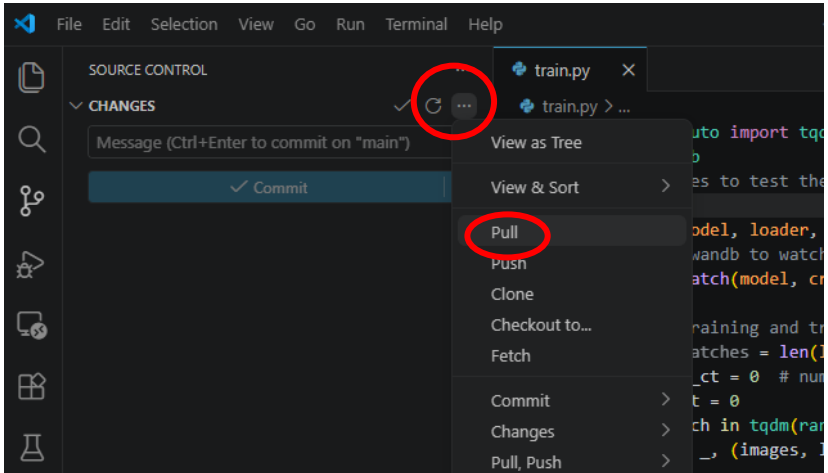
Made some changes to any file on your repo



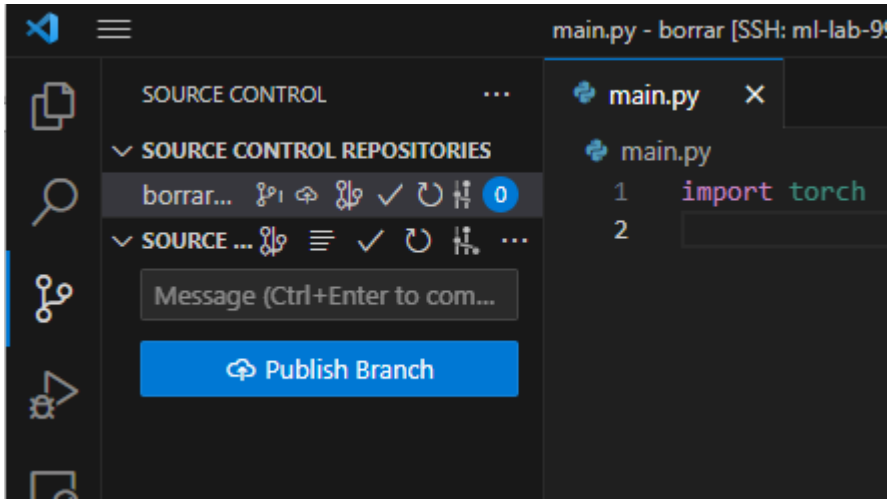
Changes detected



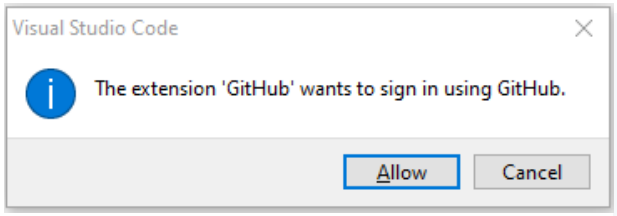


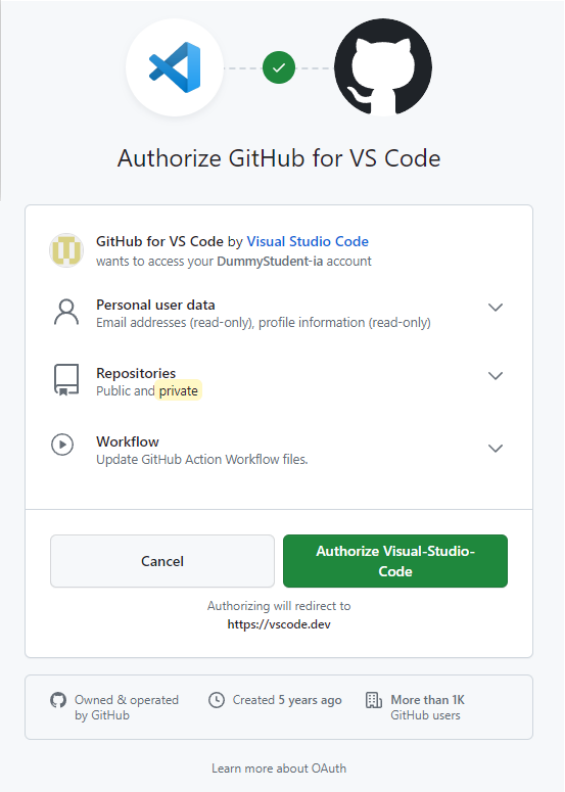


Refresh the code from your mates pulling changes when entering and regularly

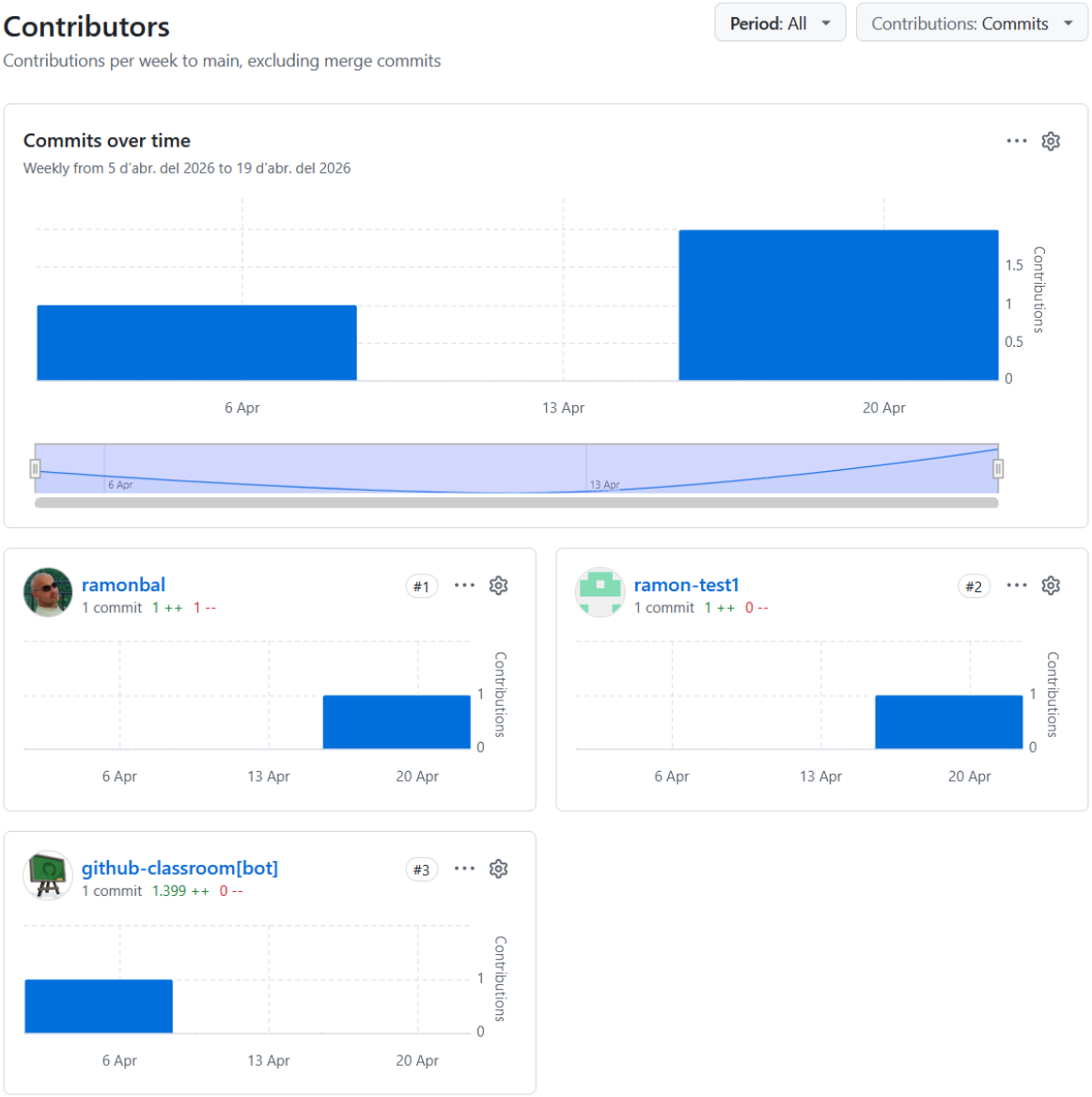


The commit you just made is local in your machine. If you want to publish this in a GitHub repo, you need to click on “Publish Branch”





If it is the first time you are using VSCode with the GitHub extension, you will be asked to authorize it to access your account



If done properly, every team member will have his/her changes logged.

Weights and Biases

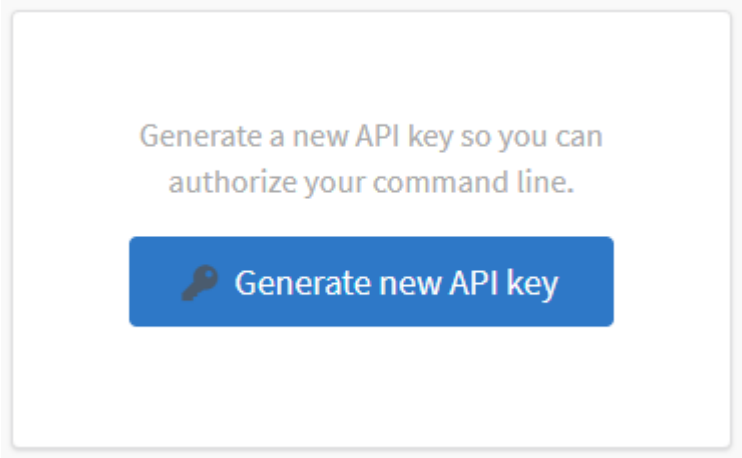
We propose that you use Weights and Biases to keep an online laboratory book with information about your training logs. Before you can use this facility, you need to login to wandb from your virtual machine. Open a terminal window and issue the following command once installed:

`pip install wandb` or `conda install -c conda-forge wandb`

```
wandb login --relogin
```

```
xnmaster@labDAGLBU:~/Projecte/XNAPproject-20$ wandb login --relogin
wandb: Logging into wandb.ai. (Learn how to deploy a W&B server locally: https://wandb.me/wandb-server)
wandb: You can find your API key in your browser here: https://wandb.ai/authorize
wandb: Paste an API key from your profile and hit enter, or press ctrl+c to quit:
```

https://wandb.ai/authorize



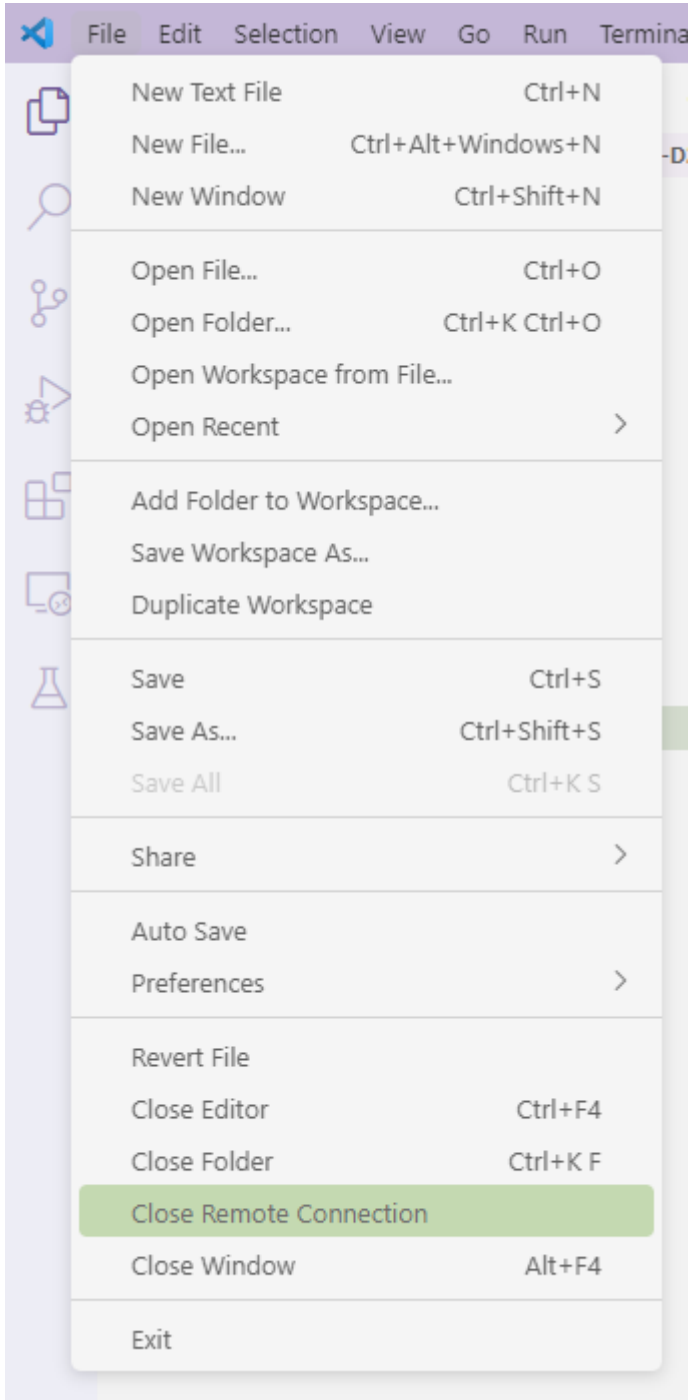
Then go to the authorization page, click on generate new API key, copy it, and paste it in the terminal

(when you paste, you will not see anything pasted, just press Ctrl+V and enter)

```
wandb: Appending key for api.wandb.ai to your netrc file: /home/xnmaster/.netrc
```

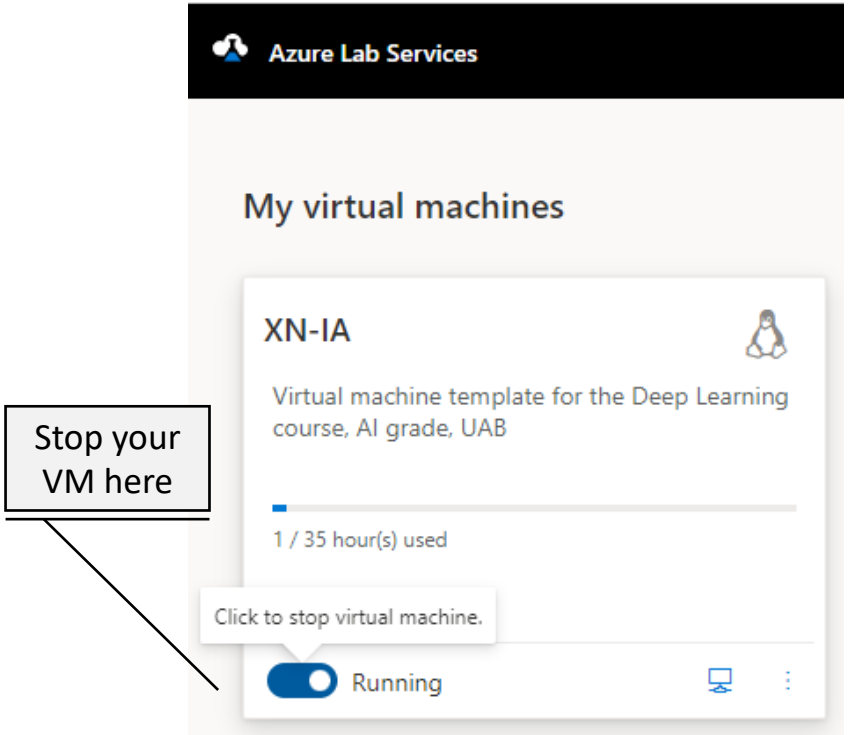
Log off!

When you finish



When you finish, close the remote connection in VSCode.

IMPORTANT:

Then go to the virtual machines and **STOP your virtual machine**. Otherwise, your time will keep counting.

https://labs.azure.com/virtualmachines